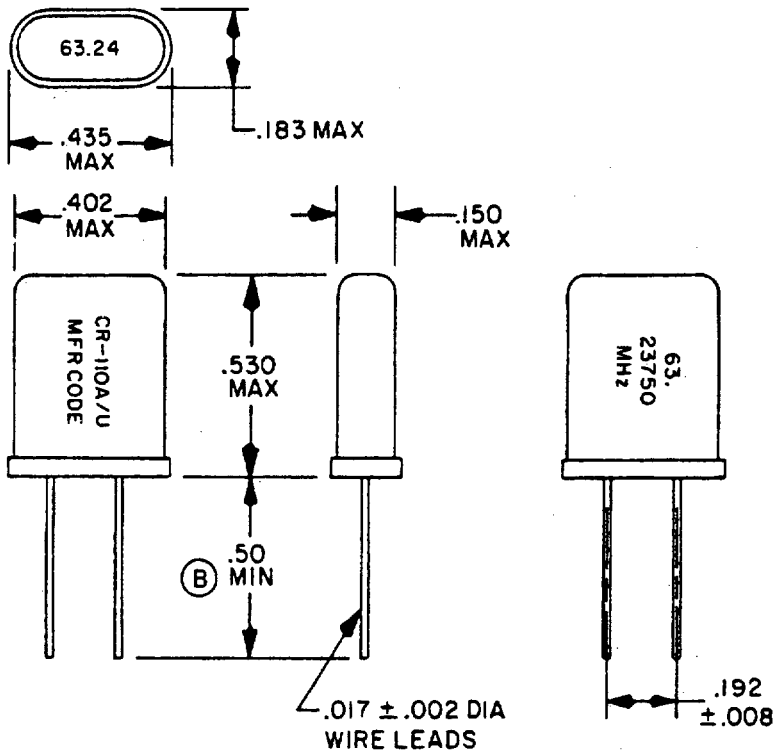
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-110A/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the crystal unit described herein shall consist of this specification and the latest issue of MIL-C-3098.

Pertinent characteristics: 62.5 to 75 MHz; fifth overtone; noncontrolled; series.



INCHES	MM
.002	.05
.008	.20
.017	.43
.150	3.81
.183	4.65
.192	4.88
.402	10.21
.435	11.05
.50	12.7
.530	13.46

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.
3. The type designator, frequency, and manufacturer's code shall be not less than .038 (.97 mm) in height and not less than .027 (.69 mm) in width.

FIGURE 1. Crystal unit - CR-110A/U.

Ⓑ denotes changes

FSC 5955

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Ⓑ TABLE I. Requirements.

Requirement	Specified value or condition
Holder	HC-18/U or HC-43/U (MIL-H-10056/9)
Dimensions, marking, and configuration	See figure 1.
Frequency range	62.5 to 75 MHz, inclusive
Capacitance, shunt	4 pF, maximum
Mode of oscillation	Fifth mechanical overtone
Government reference standard test set	TS-683/TSM
Rated drive level	2.0 ±0.4 mW
Calibration values:	
Equivalent resistance	30 ohms
Voltage drop	.25 volts
Frequency tolerances:	
Operating temperature range	±30 ppm
Reference temperature	±15 ppm
Reference temperature	25°C ±1°C
Operating temperature range (noncontrolled):	-40°C to +85°C, inclusive
Frequency tolerance	±30 ppm
Equivalent resistance	60 ohms, maximum
Unwanted modes 1/	Method II
Shock:	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Vibration: 2/	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Thermal shock:	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Aging:	
Frequency change permitted	±5 ppm

1/ Unwanted modes: The minimum unwanted mode effective resistance shall be at least two times that of the main mode, or 40 ohms, whichever is greater.

2/ Method 204 of MIL-STD-202, test condition A.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 11, 85
DLA - ES

User activities:
Air Force - 17, 84

Agent:
DLA - ES

Preparing activity:
Army - ER

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